KINGSTON
ENGINEERING COLLEGE
Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai
Accredited with 'A' Grade by NAAC.

DEPARTMENT OF SCIENCE AND HUMANITIES

SUBJECT	**: LINEAR ALGEBRA**
SUB. CODE	**: MA25C02**
YEAR / SEMESTER	**: I / II**

QUESTION BANK

UNIT III- LINEAR INNER PRODUCT
Inner product – Norms – Cauchy-schwarz inequality – Gram Schimdt orthogonalization, - Simple problems (Up to R^3).

Course Outcome: CO3

Part A

1. The standard inner product in $\mathbb{R}^3$ of (1,2,3) and (4,−1,2) is
a) 6 b) 8 c) 10 d) 4

2. The norm of (2,−1,2) is
a) 9 b) 3 c) $\sqrt{9}$ d) $\sqrt{5}$

3. If $\langle u, v\rangle = 0$, then vectors are
a) parallel b) equal c) orthogonal d) dependent

4. $\|$ (1,1,1) $\|$equals
a) 3 b) $\sqrt{2}$ c) $\sqrt{3}$ d) 1

5. Cauchy–Schwarz inequality states
a) $| u \cdot v | \geq || u || || v ||$ b) $| u \cdot v | \leq || u || || v ||$
c) $u \cdot v = || u || || v ||$ d) $u \cdot v = 0$

6. The projection of v onto u is
a) $\frac{u \cdot u}{u \cdot v} u$ b) $\frac{v \cdot v}{u \cdot u} v$ c) $\frac{u \cdot v}{u \cdot u} u$ d) $\frac{u \cdot v}{v \cdot v} v$

7. Inner product of (1,0,1) and (0,1,1) is
a) 2 b) 1 c) 0 d) −1

8. Norm of (0,0,0) is
a) 1 b) undefined c) 0 d) $\sqrt{0}$

9. Two vectors are linearly dependent if equality holds in

a) Triangle inequality
b) Gram determinant
c) Cauchy–Schwarz inequality
d) Projection formula

10. Gram–Schmidt process produces

a) independent vectors
b) orthogonal vectors
c) parallel vectors
d) zero vectors

11. The angle θ between u and v satisfies

a) $\cos\theta = \frac{u \cdot v}{\|u\|\|v\|}$
b) $\cos\theta = u \cdot v$
c) $\cos\theta = \| u \| \| v \|$
d) $\cos\theta = \frac{\|u\|}{\|v\|}$

12. $(1,-1,0) \cdot (1,1,0)$equals

a) 0
b) −2
c) 2
d) 1

13. The norm is always

a) negative
b) complex
c) non-negative
d) zero

14. In $\mathbb{R}^3$, maximum number of orthogonal non-zero vectors is

a) 1
b) 2
c) 4
d) 3

15. If u is unit vector, then $\| u \| =$

a) 0
b) 1
c) −1
d) √2

16. Inner product of perpendicular vectors is

a) 1
b) undefined
c) 0
d) −1

17. $\| (3,4,0) \|$equals

a) 7
b) 25
c) 5
d) √7

18. The projection vector is always

a) perpendicular to u
b) parallel to u
c) perpendicular to v
d) zero

19. Gram–Schmidt starts with

a) normalized vector
b) zero vector
c) first given vector
d) sum of vectors

20. $(2,0,0) \cdot (0,3,0)$equals

a) 6
b) 0
c) 3
d) 2

21. If $\| u \| = 2$and $\| v \| = 3$, maximum value of $|u \cdot v|$ is

a) 1
b) 6
c) 5
d) 9

22. Norm of (1,2,2) is

a) 3
b) √5
c) √9
d) 5

23. Orthogonal vectors have angle
a) 45° b) 60° c) 90° d) 180°

24. $(1,2,3) \cdot (0,0,1)$equals
a) 1 b) 2 c) 3 d) 0

25. Cauchy–Schwarz becomes equality when vectors are
a) orthogonal b) independent c) unit d) parallel

26. $\| (-1,-1,-1) \|$equals
a) $-\sqrt{3}$ b) $\sqrt{3}$ c) 3 d) 1

27. Inner product is
a) bilinear b) non-linear c) constant d) negative

28. In Gram–Schmidt, we subtract
a) sum of vectors b) cross product c) projection d) norm

29. Norm of (0,5,12) equals
a) 13 b) 17 c) 25 d) $\sqrt{169}$

30. If $u \cdot v = 0$ and both non-zero, vectors are
a) parallel b) identical c) orthogonal d) dependent

31. Unit vector in direction of v is
a) $v/\|v\|$ b) $\|v\|v$ c) $v \cdot v$ d) v^2

32. $(1,1,0) \cdot (1,-1,0)$equals
a) 1 b) 2 c) 0 d) -1

33. Norm squared equals
a) $\| v \|$ b) $v \cdot v$ c) $v + v$ d) $v - v$

34. Projection length equals
a) $\frac{u \cdot v}{\|u\|}$ b) $\| u \| \| v \|$ c) $u \cdot u$ d) $\frac{u \cdot v}{\|v\|}$

35. Maximum orthonormal vectors in $\mathbb{R}^2$
a) 1 b) 3 c) 4 d) 2

36. Inner product is commutative in
a) real spaces b) complex without conjugate c) none d) matrices

37. If $\|u\|=0$ then u is
a) unit b) zero vector c) orthogonal d) parallel

38. Norm satisfies
a) Triangle inequality b) Cauchy only c) Equality rule d) Zero rule

39. Gram–Schmidt converts independent vectors into

a) orthogonal basis b) zero basis c) parallel set d) equal vectors

40. (2,1,1) · (1,1,1)equals

a) 3 b) 4 c) 2 d) 5

41. If vectors are orthonormal, their norms are

a) 0 b) 2 c) 1 d) unequal

42. || (4,3,0) ||equals

a) 7 b) 5 c) 25 d) √7

43. Angle between identical vectors is

a) 0° b) 90° c) 60° d) 180°

44. Cauchy–Schwarz works in

a) only R^2 b) only R^3 c) all inner product spaces d) matrices only

45. (1,0,0) · (0,0,1)equals

a) 1 b) 0 c) −1 d) 2

46. Projection formula denominator is

a) u·v b) v·v c) u·u d) ||v||

47. Norm of (5,12,0) equals

a) 13 b) 17 c) 25 d) √169

48. Gram–Schmidt requires vectors to be

a) orthogonal b) dependent c) independent d) equal

49. If u·v > 0, angle is

a) acute b) obtuse c) 90° d) 180°

50. Inner product of (0,1,0) and (0,0,1) equals

a) 0 b) 1 c) −1 d) 2

51. || (1,0,1) ||equals

a) 1 b) 2 c) √2 d) √3

52. If vectors are parallel, projection equals

a) zero b) full vector direction c) orthogonal d) undefined

53. Gram–Schmidt output vectors are

a) normalized always b) orthogonal c) dependent d) zero

54. Norm is derived from

a) determinant b) projection c) inner product d) matrix

55. $(1,2,1) \cdot (2,1,0)$equals

a) 5 b) 4 c) 3 d) 2

56. Equality in Cauchy holds when

a) orthogonal b) zero c) parallel d) unit

57. Unit vector has norm

a) 0 b) 2 c) $\sqrt{2}$ d) 1

58. $\| (0,6,8) \|$equals

a) 10 b) 14 c) 12 d) $\sqrt{100}$

59. Projection vector lies in

a) plane b) span of u c) origin only d) null space

60. Inner product space must satisfy

a) positivity b) negativity c) singularity d) zero rule

PART-B

1. Define inner product. (K1) (A.U2018)
2. Define Frobenious inner product. (K1) (A.U2017)
3. Verify that the set of vectors {(1,0,-1), (2,0,2), (0,5,0)} is orthogonal with respect to the Euclidean inner product. (K1) (A.U2018)
4. Find the angle between the two vectors $u = (4,3,1,-2), v = (-2,1,2,3)$. (K1) (A.U2017,2025,2022)
5. Prove that the vectors $u = (1,1)$ and $v = (1,-1)$ are orthogonal with respect to the Euclidean inner product on R^2. (K2) (A.U2025)

6. Write the standard inner product space of F^n. (K1)
7. Show that $< (a, b), (c, d) > = ac - bd$ on R^2 is not an inner product. (K1)
8. Define orthogonal vectors. (K2)
9. Prove that (1,) and (1, -1) are orthogonal. (K5)
10. Define norm of a vector and unit vector. (K4)
11. Find the norm of (3, 4, 0) in R^3 with standard inner product. (K6)
12. If $\alpha = (1, 2, 3, 4) find\ ||3\alpha||$. (K5)
13. Write the formula for finding orthogonal basis using Gram-Schmidt (K3)
14. Find the norm of the following vectors in $V_3(R)$ with standard inner product: (a). (1,1,1), (b). (1,2,3) (c). $(3, -4, 0)$ (K1)
15. Show that the following function is not an inner product on R^3, where $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2 v_3)$ defined by $\langle u, v\rangle = u_1 v_1 - 2u_2 v_2 + u_3 v_3$ (K2)
16. For any x and y in an inner product space , $prove\ that\ \left||x + y|\right| \leq \left||x|\right| + \left||y|\right|$.**(K1)**
17. State Cauchy Schwarz's inequality. **(K1) (A.U2018)**
18. Define orthogonal complement. **(K1) (A.U2017)**
19. Show that the vectors $u = (-2,3,1,4)$ and $u = (1,2,0,-1)$ are orthogonal in R^1. **(K1) (A.U2022)**
20. Prove that $\left||u + v|\right|^2 + \left||u - v|\right|^2 = 2\left||u|\right|^2 + 2\left||v|\right|^2$holds for any $u, v \in V$ in an inner product space. **(K2) (A.U2024)**
21. Verify that the set of vectors {(1,0,-1),(2,0,2),(0,5,0)}is orthogonal with respect to the Euclidean inner product. **(K2) (A.U2024)**
22. Verify Cauchy Schwartz inequality for $u = \{1, -1, 3)$ and $v = (2, 0, -1)$.(K2)
23. Let $f(t) = t + 2$ and $g(t) = 3t - 2$ be the polynomials in $P(t)$and $< f, g >= \int_0^1 f(t)g(t)dt$ find $< f, g >$ (K1)
24. Let $f(t) = t + 2$ and $g(t) = t^2 - 2t - 3$ be the polynomials in $P(t)$and $< f, g >= \int_0^1 f(t)g(t)dt$ find $||f||$ and $||g||$. (K2)
25. Prove that $||x + y||^2 + ||x - y||^2 = 2||x||^2 + 2||y||^2$ for $x, y \in V$. (K1)
26. Consider the inner product space V. show that $||x + y||^2 = ||x||^2 + 2Re < x, y > + ||y||^2$ for all $x, y \in V. Re < x, y >$ denotes the real part of the complex number $< x, y >$. (K3)
27. For the inner product space V, show that $\left|\left||x|\right| - \left||y|\right|\right| \leq ||x - y||$ for all $x, y \in V$. (K3)
28. Given $u = (2, 3, 5), v = (1, -4, 3) in\ R^3$ then find the angle θ between u and v. (K2)

29. Let $f(t) = t + 2$ and $g(t) = t^2 - 2t - 3$ be the polynomials in $P(t)$and $< f, g >= \int_0^1 f(t)g(t)dt$ normalize f. (K2)

30. Let $f(t) = t + 2$ and $g(t) = 3t - 2$ be the polynomials in $P(t)$and $< f, g >= \int_0^1 f(t)g(t)dt$ normalize g. (K1)

31. Use Frobenious inner product to compute $||A||$ and $< A, B >$ for $A = \begin{pmatrix} 1 & 2+i \\ 3 & i \end{pmatrix}$ and $B = \begin{pmatrix} 1+i & 0 \\ i & -i \end{pmatrix}$. (K2)

32. Let $V = F^n$ and let $A \in M_{m\times n}(F)$ prove that $< x, Ay >=< A^*x, y >$ for all $x, y \in V$. (K1)

33. Prove that the given is norm on the given vector space V. $V = M_{m\times n}(F)$: $||A|| = \max|A_{ij}|$ $for\ all\ A \in V$. (K2)

34. $< A, B > = tr(A + B)$on $M_{2\times 2}(R)$ prove that $< A, B >$ is not a inner product on given vector space. (K1)

PART-C

1. Prove that R^2 is an inner product space with an inner product defined by $\langle x, y\rangle = a_1b_1 - a_2b_1 - a_1b_2 + 2a_2b_2$. **(K1) (A.U2019)**
2. Let $V = C^3$ with inner product $\langle x, y\rangle = a_1\overline{b_1} + a_2\overline{b_2} + a_3\overline{b_3}$ where $x = (a_1, a_2, a_3)$ and $y = (b_1, b_2, b_3)$ (i). $\langle x, y\rangle$ (ii). $||x||$ and $||y||$ (iii). $||x + y||$ (iv). Verify Cauchy and triangle inequality. **(K6) (A.U2017)**
3. Prove that R^2 is an inner product space with an inner product defined by $< x, y >= a_1b_1 - a_2b_1 - a_1b_2 + 2a_2b_2$ where $x = (a_1, a_2)$ and $y = (b_1, b_2)$. **(K1)**
4. Consider the inner product space $M_{2\times 3}(R)$ of all 2×3 matrix over R, using Frobenius inner product, compute $\langle A, B\rangle$, $||A||$, $||B||$, where
$$A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$ **(K1) (A.U2017)**
5. State and prove Cauchy Schwarz's inequality. **(K1) (A.U2019)**
6. State and Prove Gram – Schmidt Orthogonalization process. **(K1) (A.U2018)**
7. In R^4, let $v_1 = (1,0,1,0)$, $v_2 = (1,1,1,1)$, $v_3 = (0,1,2,1)$ Using Gram – Schmidt Process, compute the orthogonal vectors $w_1, w_2\ and\ w_3$ and then normalize these vectors to obtain an orthonormal set. **(K1) (A.U2016)**

8. Let R^3 have the Euclidean inner product . Use Gram-Schmidt process to transform the basis $\{u_1, u_2, u_3\}$ into orthonormal basis, $u_1 = (1,1,1)$, $u_2 = (0,1,1)$, $u_3 = (0,0,1)$ **(K2)**

9. Given that $v_1 = (1,1,1,1), v_2 = (1,1,1,0), v_3 = (1,1,0,0)$ *and* $v_4 = (1,0,0,0)$ is a basis of R^4 and assuming that we are working with the standard Euclidean inner product, construct an orthonormal basis for R^4 **(K1) (A.U2018)**

10. Find the orthogonal projection of the vector $u = (-3,-3,8,9)$ on the subspace of R^4 spanned by the vectors $v_1 = (3,1,0,1), v_2 = (1,2,1,1), v_3 = (-1,0,2,-1)$. **(K1)(A.U 2022)**

11. Assume that the vector space R^3 has the Euclidean inner product. Apply the Gram-Schmidt process to transform the basis vectors $u_1 = (1,1,1), u_2 = (0,1,1), u_3 = (0,0,1)$ into an orthogonal basis $\{v_1, v_2, v_3\}$, and then normalize the orthogonal basis vectors to obtain an orthonormal basis $\{q_1, q_2, q_3\}$. **(K2)(A.U 2023)**

12. Apply Gram-Schmidt process to transform the basis vectors $u_1 = (1,1,1)$, $u_2 = (1,1,0)$, $u_3 = (1,0,0)$ of R^3 with the inner product $\langle u, v\rangle = u_1v_1 + 2u_2v_2 + 3u_3v_3$ into orthonormal basis. **(K2)(A.U 2024)**

13. Apply Gram-Schmidt process to the given subset S of the inner product space V to obtain

    (a) Orthogonal basis for span (S)

    (b) Normalize the vectors in this basis an orthonormal basis β for span (S)

    (c) Compute the Fourier co-efficient of the given vector relative to β $V = M_{2\times2}(R), S = \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 9 \\ 5 & -1 \end{pmatrix}, \begin{pmatrix} 7 & -17 \\ 2 & -6 \end{pmatrix}$ and $A = \begin{pmatrix} -1 & 27 \\ -4 & 8 \end{pmatrix}$. (K2)

14. Apply Gram-Schmidt process to the given subset S of the inner product space V to obtain

    (a) Orthogonal basis for span (S)

    (b) Normalize the vectors in this basis an orthonormal basis β for span (S)

    (c) Compute the Fourier co-efficient of the given vector relative to β, $V = M_{2\times2}(C), S = \begin{pmatrix} -1+i & -i \\ 2-i & 1+3i \end{pmatrix}, \begin{pmatrix} -1-7i & -9-8i \\ 1+10i & -6-2i \end{pmatrix}, \begin{pmatrix} -11-132i & -34-31i \\ 7-126i & -71-5i \end{pmatrix}$ and $A = \begin{pmatrix} -7+5i & 3+18i \\ 9-6i & -3+7i \end{pmatrix}$. (K2)

15. Apply Gram-Schmidt process to the given subset $S = \{(1,0,1), (0,1,1), (1,3,3)\}$ *and* $x = (1,1,2)$ with the inner product in R^3 compute (a) Orthogonal basis for span (S)
    (b) normalize the vectors in this basis to obtain the orthonormal basis β for span (S)
    (c) Compute Fourier Co-efficient of the given vector relative to β (K2).

16. Apply Gram-Schmidt process to the given subset $S = \{(-4, 3-2i, i, 1-4i), (-1-5i, 5-4i, -3+5i, 7-2i), (-27-i, -7-6i, -15+25i, -7+6i)\} and\ x = (-13-7i, -12+3i, -39-11i, -26+5i)$ with the inner product in C^4 compute (a) Orthogonal basis for span (S)
(b) normalize the vectors in this basis to obtain the orthonormal basis β for span (S)
(c) Compute Fourier Co-efficient of the given vector relative to β. (K2)

17. Apply Gram-Schmidt process to the given subset S with the inner product space V compute (a) Orthogonal basis for span (S)
10m (b) normalize the vectors in this basis to obtain the orthonormal basis β for span (S)
$V = P_2(R)$ with inner product $< f(x), g(x) > = \int_0^1 f(t)g(t)dt,\ S = \{1, x, x^2\}$ and $h(x) = 1 + x$. (K2)

18. Apply Gram-Schmidt process to the given subset S with the inner product space V compute (a) Orthogonal basis for span (S)
(b) normalize the vectors in this basis to obtain the orthonormal basis β for span (S)
$V = span(S)$ with inner product $< f, g > = \int_0^{\pi} f(t)g(t)dt,\ S = \{\sin t, \cos t, 1, t\}$ and $h(t) = 2t + 1$. (K2)

19. Let $V = P(R)$ with inner product $< f(x), g(x) > = \int_{-1}^{1} f(t)g(t)dt$, and consider the subspace $P_2(R)$ with the standard ordered basis β. Use Gram-Schmidt process to replace β by an orthogonal basis $\{v_1, v_2, v_3\}$ for $P_2(R)$ and then use this orthogonal basis to obtain an orthonormal basis for $P_2(R)$. (K2).

UNIT IV: Matrix Decomposition

Orthogonal transformation of a symmetric matrix to diagonal form - Positive definite matrices, QR decomposition, Singular Value Decomposition (SVD), Least squares solutions- simple problems (3×3 matrices)

Course Outcome CO4

PART A

1. A real symmetric matrix can be diagonalized by

A) Any invertible matrix
B) A unitary matrix
C) An orthogonal matrix
D) A triangular matrix

2. If A is a real symmetric matrix, then its eigenvalues are

A) Complex
B) Purely imaginary
C) Zero
D) Real

3. The theorem that guarantees orthogonal diagonalization of real symmetric matrices is:

A) Cayley-Hamilton Theorem
B) Rank-Nullity Theorem
C) Spectral Theorem
D) Jordan Theorem

4. If A is symmetric, then eigenvectors corresponding to distinct eigenvalues are:

A) Orthogonal
B) Linearly dependent
C) Parallel
D) Equal

5. If P is an orthogonal matrix, then P^{-1} equals:

A) −P
B) P^T
C) P^2
D) |P|

6. Orthogonal transformation preserves:

A) Rank only
B) Determinant only
C) Inner product
D) Trace only

7. If A is symmetric and P is orthogonal, then the diagonal matrix is given by:

A) PAP^T
B) P^TAP
C) PAP
D) APP^T

8. The diagonal entries in the orthogonal diagonalization of a symmetric matrix are:

A) Eigenvalues
B) Ones
C) Zeros
D) Determinant

9. If a symmetric matrix has repeated eigenvalues, the eigenvectors:

A) Cannot be orthogonal
B) May be chosen orthogonal
C) Are always identical
D) Do not exist

10. A real symmetric matrix of order n has:

A) Exactly n real eigenvalues (counting multiplicity)
B) No real eigenvalues
C) Only one eigenvalue
D) Infinite eigenvalues

11. If A is symmetric, then A=A^T implies:

A) A is orthogonal
B) A is diagonal
C) A is normal
D) A is singular

12. Orthogonal transformation corresponds geometrically to:

A) Scaling
B) Shearing
C) Projection
D) Rotation or reflection

13. If A=PDP^T, then P consists of:

A) Row vectors of A
B) Column vectors of A
C) Eigenvectors of A
D) Diagonal elements

14. The quadratic form $x^T Ax$ can be reduced to diagonal form using:

A) Similarity transformation
B) Orthogonal transformation
C) Row operations
D) Elementary operations

15. Orthogonal matrices satisfy:

A) $PP^T = I$
B) $PP = I$
C) $P^T = -P$
D) $P^2 = 0$

16. If AAA is symmetric positive definite, then its eigenvalues are:

A) Negative
B) Complex
C) Positive
D) Zero

17. Two symmetric matrices are orthogonally similar if:

A) They have the same rank
B) They have the same eigenvalues
C) They are equal
D) Their determinants are equal

18. The trace of a symmetric matrix equals

A) Product of eigenvalues
B) Sum of eigenvalues
C) Zero
D) Rank

19. Determinant of a symmetric matrix equals

A) Sum of eigenvalues
B) Product of eigenvalues
C) Trace
D) Zero

20. The process of orthogonal diagonalization involves

A) Finding eigenvalues and orthonormal eigenvectors
B) Row reduction only
C) Matrix inversion
D) Cofactor expansion

21. The QR decomposition of a matrix A expresses A as:

A) $A=LU$
B) $A= Q\Lambda Q^T$
C) $A= QR$
D) $A=U \Sigma V^T$

22. In QR decomposition, matrix Q is:

A) Upper triangular
B) Orthogonal
C) Diagonal
D) Symmetric

23. Matrix R in QR decomposition is:

A) Orthogonal
B) Lower triangular
C) Upper triangular
D) Diagonal

24. If Q is orthogonal, then:

A) $Q^TQ = I$
B) $QQ = I$
C) $Q^2 = Q$
D) $Q^{-1} = Q$

25. QR decomposition is commonly used to:

A) Compute determinants
B) Solve least squares problems
C) Find eigenvalues directly
D) Compute matrix inverse only

26. QR decomposition can be computed using:

A) Gaussian elimination
B) Gram–Schmidt process
C) Laplace expansion
D) Cramer's rule

27. If A is an m×n matrix with m > n, then Q is:

A) m×m
B) n×n
C) m×n
D) n×m

28. Reduced QR decomposition produces:

A) Square Q always
B) Smaller orthogonal Q
C) Diagonal Q
D) Identity Q

29. QR factorization exists for:

A) Only square matrices
B) Only symmetric matrices
C) Any matrix
D) Only invertible matrices

30. In least squares problem Ax = B, QR decomposition helps because:

A) Q is diagonal
B) R is triangular and easy to solve
C) A becomes symmetric
D) It reduces rank

31. The Singular Value Decomposition of a matrix A is:

A) $A = QR$
B) $A = LU$
C) $A = U\Sigma V^T$
D) $A=PDP^{-1}$

32. In SVD, U is:

A) Diagonal
B) Orthogonal
C) Upper triangular
D) Symmetric

33. Matrix Σ in SVD contains:

A) Eigenvalues
B) Singular values
C) Determinants
D) Pivots

34. Singular values are:

A) Always negative
B) Always positive or zero
C) Complex numbers
D) Equal to eigenvalues always

35. SVD exists for:

A) Only square matrices
B) Only invertible matrices
C) Any real or complex matrix
D) Only symmetric matrices

36. The singular values of A are square roots of eigenvalues of:

A) A
B) A^{-1}
C) A^TA
D) $A + A^T$

37. The rank of matrix A equals:

A) Number of zero singular values
B) Number of nonzero singular values
C) Trace of A
D) Determinant of A

38. SVD is widely used in:

A) Cryptography only
B) Image compression
C) Solving quadratic equations
D) Polynomial division

39. In SVD, matrices U and V are:

A) Diagonal
B) Orthogonal (or unitary in complex case)
C) Triangular
D) Identity

40. The condition number of matrix A using SVD is:

A) Largest singular value
B) Smallest singular value
C) Ratio of largest to smallest singular value
D) Sum of singular values

41. The least squares solution of Ax=b is used when:

A) A is invertible
B) The system is inconsistent
C) A is diagonal
D) b=0

42. The least squares solution minimizes:

A) $\|x\|$
B) $\|Ax\|$
C) $\|Ax-b\|$
D) $\|A-b\|$

43. The normal equations are:

A) $Ax=b$
B) $A^TAx = A^Tb$
C) $AA^Tx = b$
D) $A^{-1}b$

44. For the least squares solution to be unique, the columns of A must be:

A) Orthogonal
B) Linearly dependent
C) Linearly independent
D) Equal

45. If A is m × n, then A^TA is:

A) $m \times m$
B) $n \times n$
C) $m \times n$
D) $n \times m$

46. The vector $A\hat{x}$ (least squares solution) represents:

A) Projection of b onto column space of A
B) Projection of b onto null space
C) Zero vector
D) Row space vector

47. The residual vector is:

A) $x - b$
B) Ax
C) $b - A\hat{x}$
D) $A^T b$

48. The residual vector is orthogonal to:

A) Null space of A
B) Column space of A
C) Row space of A
D) b

49. If columns of A are orthonormal, least squares solution is:

A) $A^{-1} b$
B) $A^T b$
C) $A A^T b$
D) $A^T A b$

50. In data fitting, least squares minimizes:

A) Sum of errors
B) Sum of squared errors
C) Maximum error
D) Absolute error

51. The least squares solution exists when:

A) $A^T A$ is invertible
B) A is square
C) $b = 0$
D) A is identity

52. If A has full column rank, then:

A) $A^T A$ is singular
B) $A^T A$ is invertible
C) $A^T = A$
D) A is orthogonal

53. The least squares solution can be written as:

A) $(A^T A)^{-1} A^T b$
B) $A(A^T A)^{-1} b$
C) $A^{-1} b$
D) $AA^T b$

54. If the system is consistent, the least squares solution:

A) Is different from exact solution
B) Equals exact solution
C) Is zero
D) Does not exist

55. In geometric terms, least squares finds:

A) Closest vector in column space to b
B) Largest vector
C) Smallest column
D) Null space vector

56. QR factorization helps compute least squares by:

A) Avoiding normal equations
B) Making matrix singular
C) Squaring matrix
D) Eliminating variables randomly

57. If A=QRA = QRA=QR, then least squares solution satisfies:

A) $Rx = Q^T b$
B) $Qx = Rb$
C) $Ax = Q b$
D) $Rx = b$

58. The projection matrix onto column space of A is:

A) $A^T(AA^T)^{-1} A$
B) $A(A^TA)^{-1}A^T$
C) $(A^TA)^{-1}$
D) AA^T

59. The least squares method is widely used in:

A) Curve fitting
B) Cryptography only
C) Sorting
D) Encryption only

60. If A^TA is not invertible, we may use:

A) Identity matrix
B) Pseudoinverse
C) Zero matrix
D) Determinant method

PART B

1. Define orthogonal transformation. **(K1)**
2. Define positive definite, negative definite. **(K1)**
3. Define positive semi-definite and positive semi-definite. **(K1)**
4. Verify whether the matrix of the quadratic form $11x^2 + 2y^2 + 2z^2 + 4xy - 2yz + 4xz$ is positive definite nor not. **(K4)**
5. Verify whether the matrix of the quadratic form $2x^2 + 2xy + 3z^2$ is positive semi-definite nor not. **(K4)**
6. Find the nature of the matrix of the quadradic form $x_1^2 + x_2^2$. **(K1)**
7. Find the nature of the matrix of the quadradic form $2x_1^2 + 5x_2^2 + 3x_3^2 + 4x_1x_2$. **(K1)**
8. Find the nature of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$. **(K1)**
9. Verify that $A = A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ is a positive definite. **(K4)**

10. Verify that A = A = $\begin{pmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{pmatrix}$ is a positive definite. **(K4)**

11. Test AA^T is positive definite if A = $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. **(K4)**

12. Test the matrix A = $\begin{pmatrix} 1 & 1+i \\ 1-i & 2 \end{pmatrix}$ is a positive definite. **(K4)**

13. Write the matrix of the quadratic form $2x^2+8z^2+4xy+10xz-2yz$. **(K2)(A.U 2013)**

14. Write the matrix of the quadratic form $2x_1^2 - 2x_2^2 + 4x_3^2 + 2x_1x_2 - 6x_1x_3 + 6x_2x_3$. **(K2)(A.U 2022)**

15. Write the quadradic form to the symmetric matrix $\begin{pmatrix} 2 & 0 & -2 \\ 0 & 2 & 1 \\ -2 & 1 & -2 \end{pmatrix}$. **(K2)(A.U 2013)**

16. Write the quadradic form to the symmetric matrix $\begin{pmatrix} 1 & 2 & 3 \\ 2 & -2 & -4 \\ 3 & -4 & -3 \end{pmatrix}$. **(K2)(A.U 2012)**

17. Define singular value decomposition of a matrix. **(K1)**

18. Define least-square solution. **(K1)**

PART C

1. Reduce the symmetric matrix $\begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$ to a diagonal matrix by orthogonal transformation. **(K2)(A.U 2006)**

2. Diagonalize matrix $\begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ by orthogonal transformation. **(K2)(A.U 2006)**

3. Reduce the quadratic form $10x_1^2 + 2x_2^2 + 5x_3^2 + 6x_2x_3 - 10x_3x_1 - 4x_1x_2$ to a canonical form. Discuss its nature. **(K2)(A.U 2021)**

4. Diagonalise the matrix A = $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$. **(K2)(A.U 2011, 2014)**

5. Diagonalise the matrix A = $\begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$. **(K2)(A.U 2012)**

6. Decide for or against the positive definiteness of A = $\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$. **(K2)(A.U 2003, 2007, 2010, 2012,2014)**

7. Find QR decomposition of A = $\begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}$ by Gram-schmidt method. **(K1)**

8. Find QR decomposition of A = $\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ by Gram-schmidt method. **(K1)**

9. Find QR decomposition of $A = \begin{bmatrix} 2 & -2 & 18 \\ 2 & 12 & 0 \\ 1 & 2 & 0 \end{bmatrix}$ by Householder method. **(K1)**

10. Find the singular value decomposition of $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}$. **(K1)**

11. Find the SVD of $A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$. **(K1)**

12. Find the least-squares solution of AX = b for $A = \begin{bmatrix} 1 & -3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$ by using QR decomposition. **(K1)**

13. Find the least-squares solution of AX = b for $A = \begin{bmatrix} 2 & -2 & 18 \\ 2 & 12 & 0 \\ 1 & 2 & 0 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$. **(K1)**

14. Find the least square solution of the system of equations
$2x + 3y + 2z = 25$, $2x - y + 3z = 30$, $3x + 4y - 2z = 20$, $3x + 5y + 4z = 55$. **(K1)**

15. Find the least square solution of the system of equations
$2x + y = 6$, $x + y = 8$, $-2x + y = 11$, $-x + y = 8$ and $3x + y = 4$. **(K1)**

16. Fit a second degree parabola for the data by the method of least square. **(K1)**

X	0	1	2	3	4
Y	1	0	3	10	21

17. Fit a second degree parabola for the data by the method of least square. **(K1)**

X	-2	-1	0	1	2
Y	15	1	1	3	19

Prepared by	Verified by	Recommended by
3/3/26 **FACULTY INCHARGE**	3/3/26 **HOD S&H**	**PRINCIPAL**